

When Does Retrieved Text Improve Sequential Decisions?

Consequence-Aware Evaluation of Evidence Selection

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Abstract. Retrieval is normally evaluated by relevance and downstream generation quality. Many operational deployments instead consume retrieved text through a decision policy, where the cost of an error depends on which action it induces and on how the system evolves afterwards. We present an executable benchmark that measures the value of retrieved evidence by the sequential utility of the decisions it produces, holding the interpreter and controller fixed. On an operational report feed with entity-scoped distractors, stale reports and routine traffic, we compare BM25, TF-IDF and dense retrieval against random and oracle evidence selection under matched evidence budgets, and pair each system with a fixed probabilistic interpreter and a receding-horizon inventory controller. Retrieval quality and decision utility do not induce the same ordering. With one interpreter, every practical retrieval system is worse than retrieving nothing while the oracle evidence selector is clearly better, and improving nDCG can lower realised reward; with another interpreter the two orderings agree exactly. The mechanism is that non-relevant documents are not equally harmful — a current report about a different site moves the controller the wrong way, whereas an off-topic memo only wastes an evidence slot — so utility is a property of the pipeline rather than of the ranking. We also show that text-free controls are a demanding baseline: a constant belief tuned only on development seeds matches or exceeds most text interpreters, and in one transfer a shuffled-text ablation is no worse than the genuine interpreter, so positive effects against an uninformed prior do not by themselves establish the value of reading text. The benchmark, its frozen artifacts and its controls are released to support consequence-aware evaluation of information access.

Keywords: Evaluation · Evidence selection · Downstream utility · Retrieval-augmented decision making · Calibration.

1 Introduction

An operator monitoring a supply network receives a stream of reports. Most concern other sites; some are stale; a few describe conditions that should change what they order this period. Choosing which reports to act on is an information access problem, and its consequences are sequential: an order placed now arrives several periods later, into a state that earlier decisions produced.

Standard retrieval evaluation scores such a system by topical relevance, and downstream-aware evaluation such as eRAG scores it by the quality of the text a model subsequently generates [17]. Neither captures what the operator is actually exposed to. Two rankings with identical nDCG can carry different operational risk, because the ways a document can be non-relevant are not interchangeable: a current report about a *different* site, written in the same operational register but describing a different regime, will move a controller in the wrong direction; an off-topic notice merely consumes an evidence slot.

We ask a single question: *when does retrieved text improve sequential decisions?* To answer it we build an executable path from corpus to consequence,

$$q, \mathcal{D} \longrightarrow R_k(q, \mathcal{D}) \longrightarrow b(z \mid \cdot) \longrightarrow \pi(a_t \mid s_t, b) \longrightarrow s_{t+1} \longrightarrow U, \quad (1)$$

in which a retrieval system R_k selects k documents from a candidate pool $\mathcal{D}$ under a fixed query q , a fixed interpreter maps the selected text to a belief b over operational regimes, a fixed controller maps state and belief to an action, and a simulator returns realised utility U . Because every stage after retrieval is held constant, differences in U are attributable to evidence selection.

Contributions. (i) A consequence-aware evaluation of evidence selection: retrieval is scored by the sequential decision utility it induces, reported alongside conventional retrieval metrics on the same episodes (Section 4). (ii) Evidence that the two orderings diverge, with a mechanism: non-relevant documents differ in how they mislead, so equal nDCG does not imply equal utility. (iii) Demanding text-free controls — constant-belief and development-tuned no-text operating points, shuffled-text and label-only ablations — strong enough to overturn naive readings of positive effects against an uninformed prior (Section 5). (iv) A reproducible artifact: frozen corpora, embeddings, checkpoints and semantic outputs, reconstructible offline without credentials, with the estimand, weighting and uncertainty declared.

Scope. The environments and report text are synthetic. We do not introduce a retrieval model, and we do not claim that any interpreter or controller is universally preferable. The claim is about *evaluation*: an information-access system that consumes text through a decision policy should be scored on the decisions, and the controls that make such a claim credible are stronger than those typically reported.

2 Related work

Downstream-aware retrieval evaluation. eRAG evaluates retrieval by the downstream performance of the language model that consumes it, showing that upstream relevance can be a poor proxy for end-task utility [17]. CUE-R estimates per-evidence utility by intervention and includes no-retrieval controls, and GroGU makes grounding utility explicit as a consumer-specific quantity; both

are recent preprints and we treat them as such [2,7]. Our downstream consumer is a control policy in a stochastic environment rather than a generator, so the estimated quantity is a trajectory’s realised return, not text quality, and the error asymmetry comes from the environment’s cost structure rather than a judge. That relevance is situational and task-dependent long predates this work [1]; our contribution is an executable instrument for one consequential task, not the observation that consequences matter.

Decision-focused prediction and calibration. Predict-then-optimize and decision-focused learning train or evaluate predictions through a downstream objective [4,5,13]; value-aware model learning makes the analogous point for learned dynamics [6,12], decision calibration formalises the dependence on a decision-maker [18], and value-of-information theory gives the classical account of when information changes a decision [9]. Probability quality is standardly assessed with proper scores and calibration diagnostics [8,3,14]. We do not learn through the objective: we hold the pipeline fixed and vary the information supplied to it, which is what makes this an evaluation rather than a training method. We also separate calibration from the ensembling that accompanies it in common implementations (Section 6).

Environment and baselines. The multi-echelon simulator derives from an inventory benchmark lineage in operations research [10]. The retrieval baselines are Okapi BM25 [16] and cosine similarity over sentence embeddings [15], scored with nDCG [11].

3 Framework

3.1 Objects and estimand

Let z be a latent operational regime, $\mathcal{D}$ a candidate pool of reports and q a fixed query expressing a standing information need. A retrieval system returns an ordered list; the top k documents, R_k , are the evidence. A fixed interpreter maps their concatenation to a belief $b(z)$, and a fixed controller maps state and belief to an action.

Write $G_i(m)$ for the *realised* return of episode i under information source m ,

$$G_i(m) = \sum_{t=0}^{T-1} r\left(s_t^{(i)}, a_t^{(i)}, z^{(i)}\right), \quad (2)$$

undiscounted: the horizon is finite and fixed at $T = 40$, and any discounting applied inside the environment is part of r . We keep realised returns distinct from their expectation, which we estimate by the weighted benchmark return

$$J_w(m) = \sum_r w_r \operatorname{mean}_{f \in \mathcal{F}_r} \operatorname{mean}_{v \in \mathcal{V}_{r,f}} \operatorname{mean}_{i \in \mathcal{S}} G_i(m; r, f, v), \quad (3)$$

averaging seeds within a template variant v , variants within an ambiguity family f , families within a regime r , and combining regimes with declared weights w_r . The *balanced* estimand uses uniform w_r and is primary; a deployment-prior estimand using declared regime frequencies is reported as secondary. Group sizes are unequal, so (3) is not an equal-row mean over episodes, and the distinction is numerically material.

For a no-information reference n and an oracle-belief reference o , the oracle information value is $\text{OIV} = J_w(o) - J_w(n)$, and the normalised diagnostic is

$$\text{SIVR}(m) = \frac{J_w(m) - J_w(n)}{J_w(o) - J_w(n)}, \quad \text{same } w_r \text{ in numerator and denominator.} \quad (4)$$

SIVR is secondary throughout: it is undefined when the denominator is numerically negligible, flagged when $\text{OIV} < 0$, and not bounded above by one, because o is a perfect *belief* passed through a fixed controller rather than an optimal policy. Raw paired effects and their intervals come first. The implementation’s 10^{-9} guard detects a numerically zero denominator only; whether OIV is statistically distinguishable from zero is a separate question, answered by the interval.

3.2 Uncertainty

The design crosses worlds (seeds) with language (templates): every template within a regime is evaluated on the same seeds, so seeds are not independent draws nested under templates. We resample the two axes independently — one *shared* seed sample per regime, reused by every family and variant, plus families and variants with replacement. Intervals are percentile intervals from B replicates; p -values are centred bootstrap test inversions, $(1 + \#\{|b^* - \hat{\theta}| \geq |\hat{\theta}|\})/(B + 1)$, bounded below by $1/(B + 1)$ and never reported as exactly zero. With three designed ambiguity families per regime the family axis is small, so intervals are conditional on the fixed template set and generalisation to new language remains open.

4 Experiment R: retrieval quality versus decision utility

4.1 Setup

Each episode is a (seed, regime) pair. The candidate pool contains four kinds of document, all in the same operational register:

- relevant** a current report about the operator’s own node;
- wrong entity** a current report about a *different* node, drawn across all regimes, so some of these errors are actively misleading and some are accidentally correct;
- stale** the operator’s node, but an event already marked resolved;
- off-topic** routine corporate traffic attributed to the operator’s node, so it cannot be filtered on entity alone.

Table 1. Experiment R at a matched evidence budget $k = 3$, with the calibrated TF-IDF interpreter and the receding-horizon controller held fixed. Left: ranking quality. Right: what the interpreter actually receives and what the decisions are worth. ‘Misleading’ counts retrieved documents that are non-relevant *and* describe a regime other than the true one — the errors that move the controller, as opposed to those that only waste a slot. ΔJ is the paired effect against the no-retrieval control; intervals are crossed bootstrap percentile intervals with shared seed draws.

System	Ranking quality			Consequences		
	nDCG@3	R@3	MRR	Misleading	Belief acc.	ΔJ (95% CI)
No retrieval	0.000	0.000	0.000	0.00	0.333	+0.0 [+0, +0]
Random	0.090	0.117	0.218	1.33	0.367	-49.8 [-78, -19]
BM25	0.274	0.294	0.444	1.28	0.433	-74.3 [-99, -48]
TF-IDF cosine	0.336	0.356	0.517	0.97	0.456	-61.1 [-88, -32]
Dense (MiniLM)	0.617	0.672	0.727	0.87	0.589	-1.5 [-31, +29]
Oracle selector	1.000	1.000	1.000	0.67	0.756	+69.0 [+40, +96]

Relevance is determined by the query — entity and recency — and never by regime; identifying the regime is the interpreter’s job. This separation is what makes the retrieval task solvable and the dissociation measurable.

Systems: random (floor), BM25, TF-IDF cosine, dense sentence-embedding cosine, and an oracle evidence selector (ceiling). A no-retrieval control that reads nothing supplies the paired reference. Evidence budgets $k \in \{1, 3, 5\}$ are matched across systems. Interpreters are the rule-based extractor and the calibrated TF-IDF classifier; the controller is the receding-horizon optimiser.

4.2 Findings

The retrieval task is solvable, and the systems separate on it. Ranking quality orders the systems random < BM25 < TF-IDF cosine < dense < oracle, with a wide spread in nDCG@3 (Table 1). Belief accuracy rises monotonically with nDCG for both interpreters, so the upstream half of the pipeline behaves as intended: better evidence does produce better beliefs.

Better beliefs do not imply better decisions. The downstream half does not follow. With the calibrated TF-IDF interpreter at $k = 3$, *every* practical retrieval system is worse than retrieving nothing: BM25 loses 74.3 reward against the no-retrieval control (95% CI [-99, -48]) despite tripling random’s nDCG, TF-IDF cosine loses 61.1, and only dense retrieval reaches parity (-1.5, [-31, +29]). The oracle selector, by contrast, gains 69.0 ([+40, +96]). Ranking quality and decision utility therefore agree only weakly: Kendall’s τ_b between the two orderings over the four ranking systems is +0.33 at $k = 3$ and 0.00 at $k = 1$ and $k = 5$ (Table 2). An evaluation that stopped at nDCG would report steady progress across a region in which realised utility is flat or negative.

Table 2. Experiment R. Do retrieval quality and decision utility rank the ranking systems the same way? Kendall’s τ_b is computed between the nDCG@ k ordering and the J_w ordering over {random, BM25, TF-IDF, dense}; the no-retrieval and oracle references are excluded because neither is a retrieval system. The last two columns track the strongest ranker across budgets: raising its nDCG does not raise its realised effect.

Interpreter	k	τ_b	Best system by		Dense	
			nDCG	reward	nDCG@ k	ΔJ
RuleBased	1	+0.67	Dense (MiniLM)	TF-IDF cosine	0.556	+21.3
RuleBased	3	+1.00	Dense (MiniLM)	Dense (MiniLM)	0.617	+48.4
RuleBased	5	+1.00	Dense (MiniLM)	Dense (MiniLM)	0.691	+46.0
TF-IDF calib.	1	+0.00	Dense (MiniLM)	Random	0.556	-69.3
TF-IDF calib.	3	+0.33	Dense (MiniLM)	Dense (MiniLM)	0.617	-1.5
TF-IDF calib.	5	+0.00	Dense (MiniLM)	Dense (MiniLM)	0.691	-18.9

The gap has a mechanism, and it is not noise. Non-relevant documents are not interchangeable. The right half of Table 1 counts what each system actually puts in front of the interpreter: retrieved documents that are non-relevant *and* describe a regime other than the true one. Random retrieval surfaces 1.33 such documents per episode and BM25 1.28 — BM25 raises nDCG by preferring operationally-worded reports, but many of those reports are current reports about a *different* node, which is precisely the error that moves the controller. Dense retrieval improves because it reduces misleading evidence to 0.87, not merely because it ranks relevant documents higher. Standard graded relevance treats an off-topic notice and a confident report about another site as equally non-relevant; the controller does not.

Whether the dissociation appears depends on the consumer. With the rule-based interpreter the two orderings agree ($\tau_b = 1.0$ at $k = 3$ and $k = 5$) and every retrieval system beats the no-retrieval control. The same retrieval systems, the same corpus and the same controller yield opposite practical conclusions under two different interpreters, because the rule-based extractor degrades gracefully on mixed evidence while the calibrated classifier becomes confidently wrong. Utility is a property of the pipeline, not of the ranking alone, which is the concern GroGU raises for grounding utility [7] and which we can here measure end to end.

More evidence is not better. The last two columns of Table 2 follow the strongest ranker across budgets. Moving from $k = 3$ to $k = 5$ raises dense retrieval’s nDCG (0.617 $\rightarrow$ 0.691) while lowering its realised effect under the calibrated interpreter (−1.5 $\rightarrow$ −18.9). The additional documents are mostly non-relevant, and the interpreter has no mechanism for discounting them.

Table 3. Experiments M (held-out multi-echelon transfer, demand surge) and S (supply-side capacity drop). Rows are grouped as text-free controls, text interpreters, degraded-text controls, and the oracle-belief reference. J_w is the balanced weighted return and ΔJ the paired effect against NoInfo, with crossed bootstrap intervals. The dev-tuned no-text row is the one every interpreter must beat, and in both experiments most do not.

Information source	Experiment M			Experiment S		
	J_w	ΔJ	95% CI	J_w	ΔJ	95% CI
NoInfo (prior)	467.7	+0.0	[+0, +0]	1333.6	+0.0	[+0, +0]
Constant $p_{\text{event}} = 0$	381.2	-86.5	[-93, -80]	1166.7	-166.9	[-167, -167]
Constant $p_{\text{event}} = 0.5$	501.1	+33.3	[+32, +35]	1333.6	+0.0	[+0, +0]
Constant $p_{\text{event}} = 1$	560.4	+92.7	[+90, +96]	1725.1	+391.5	[+387, +396]
No-text (dev-tuned)	560.4	+92.7	[+90, +96]	1725.1	+391.5	[+387, +396]
RuleBased	568.7	+101.0	[+95, +106]	1447.0	+113.4	[+39, +183]
TF-IDF (single)	516.2	+48.5	[+36, +59]	1362.8	+29.2	[-4, +68]
TF-IDF (calib., isotonic)	530.2	+62.5	[+5, +96]	1372.9	+39.3	[-73, +131]
GPT-4o	534.9	+67.1	[+29, +87]	1372.6	+38.9	[-39, +85]
TF-IDF cal. (label only)	523.5	+55.8	[-23, +89]	1392.0	+58.4	[-73, +115]
TF-IDF cal. (shuffled)	451.1	-16.6	[-66, +52]	1415.2	+81.6	[-64, +258]
Oracle belief	552.0	+84.3	[+77, +91]	1446.3	+112.7	[+109, +116]

5 Text-free and degraded-text controls

Experiment R measures evidence selection against a no-retrieval reference. A second question is what the *text* contributes at all, once a competitive text-free policy is allowed. We evaluate two transfer environments — a held-out multi-echelon demand-surge setting (Experiment M) and a structurally different supply-side capacity drop (Experiment S) — against constant-belief endpoints, a no-text operating point tuned only on development seeds, a label-only ablation, and a shuffled-text ablation in which the interpreter reads a real warning belonging to a different template.

A constant belief is a demanding baseline. In Experiment M the development-tuned no-text control selects $p_{\text{event}} = 1$ and returns 560.4, beating single TF-IDF (516.2), calibrated TF-IDF (530.2) and GPT-4o (534.9), and exceeding the oracle-belief reference itself (552.0). Only the rule-based interpreter (568.7) is above it. In Experiment S the same control returns 1725.1 against 1447.0 for the best interpreter and 1446.3 for the oracle-belief reference — more than three times the oracle information value, without reading anything. Every interpreter’s effect against the uninformed prior is positive in both experiments; in Experiment S only RuleBased’s interval excludes zero, and none of them establishes value over competitive text-free operation. The mechanism is visible in the tuning grid: reward increases monotonically in p_{event} , so the base-stock controller is

systematically under-positioned under the benchmark prior, and any belief that raises the order-up-to level captures part of that gap without reading anything.

The shuffled-text control separates the two experiments. In Experiment M, shuffling text destroys the effect: the calibrated interpreter falls from 530.2 to 451.1, below the uninformed prior. Reading the *right* text is doing real work there. In Experiment S it does not: the shuffled control returns 1415.2, *above* the genuine interpreter’s 1372.9, even though its belief accuracy is roughly half. Whatever gain the calibrated interpreter shows in Experiment S comes from the shape of its belief distribution moving the controller toward higher stock, not from identifying the regime. We report this as a negative result: the capacity-drop transfer does not support a claim that reading the warning is what produced the improvement. Its wide interval $([-64, +258])$ also shows how little the frozen template set constrains that arm.

Label-only beliefs. Collapsing the calibrated interpreter to its argmax costs little in Experiment M ($530.2 \rightarrow 523.5$) and helps in Experiment S ($1372.9 \rightarrow 1392.0$). Probability quality, as measured by proper scores, is therefore not the operative variable in these transfers, matching what the controlled benchmark shows (Section 6).

6 Calibration is not an isolated intervention

A recurring claim in this literature is that recalibrating a classifier improves downstream value even when it costs accuracy. The controlled benchmark (Experiment C) lets us test that claim carefully, and the care turns out to matter.

The usual comparison is confounded. `CalibratedClassifierCV` does not apply a probability map to a fixed model: with k -fold cross-validation it fits k classifiers and averages them. A raw-versus-calibrated contrast therefore mixes a calibration map with an ensembling change. Adding the missing arm — the same fold ensemble with no calibration map — separates them. Ensembling alone *lowers* realised value (SIVR $0.245 \rightarrow 0.154$), and the calibration map then recovers more than it lost ($0.154 \rightarrow 0.338$). The conclusion that calibration helps survives, but the naive $0.245 \rightarrow 0.338$ difference is not attributable to calibration, and reporting it as such overstates what was shown.

Small calibration sets. With three folds over eighteen training texts each calibration set holds six points. Isotonic regression saturates there, driving some in-sample probabilities to exactly one; Platt scaling does not. We therefore report both, and note that the sigmoid arm has higher accuracy (0.861 versus 0.806) but lower realised value (0.261 versus 0.338). Accuracy costs are also larger than previously reported once the fold vocabulary is fitted inside each fold rather than on the full training set beforehand: three held-out texts of thirty-six, not one.

Table 4. Experiment C (controlled benchmark). Calibration is not an isolated intervention: **CalibratedClassifierCV** refits one classifier per fold, so the calibrated arm differs from the single-model arm in two ways. The fold-ensemble row carries the ensembling change without a calibration map and is the control that separates them. Isotonic regression on six calibration points per fold saturates; the sigmoid arm is the small-sample alternative. Effects are against the uninformed prior, which returns 1635.

Interpreter	Acc.	Brier	Log-loss	ΔJ	95% CI	SIVR
NoInfo (prior)	0.333	0.668	1.101	+0.0	[+0, +0]	0.000
No-text (uniform)	0.333	0.667	1.099	+12.5	[+11, +14]	0.047
No-text (dev-tuned)	0.333	1.333	15.351	+194.8	[+169, +221]	0.737
RuleBased	0.833	0.319	1.111	+205.7	[+161, +245]	0.779
TF-IDF (single)	0.889	0.552	0.932	+64.8	[+48, +81]	0.245
TF-IDF (label only)	0.889	0.222	2.558	+204.3	[+85, +275]	0.773
TF-IDF (fold ensemble)	0.889	0.583	0.977	+40.6	[+23, +60]	0.154
TF-IDF (calib., isotonic)	0.806	0.319	0.542	+89.4	[+13, +159]	0.338
TF-IDF (calib., sigmoid)	0.861	0.392	0.699	+69.0	[+26, +109]	0.261
TF-IDF cal. (label only)	0.806	0.389	4.477	+142.3	[+0, +265]	0.539
TF-IDF cal. (shuffled)	0.417	0.849	3.716	-16.5	[-100, +72]	-0.062
Oracle belief	1.000	0.000	0.000	+264.2	[+238, +289]	1.000

Probability quality is not the operative variable here. The strongest single result in this experiment is negative for the standard story. Collapsing the raw classifier to its hard label — discarding all probabilistic information — raises SIVR from 0.245 to 0.773, more than either calibrated arm achieves. The label-only ablation has the same accuracy as the model it is derived from, a better Brier score (0.222 versus 0.552) and a much worse log-loss (2.558 versus 0.932); the two proper scores disagree about it, and neither predicts its operational advantage. What the controller responds to is where the belief mass sits relative to its own asymmetric costs, which no upstream score measures.

And the no-text control is above all of them. The development-tuned constant belief reaches SIVR 0.737, above every TF-IDF arm and just below the rule-based extractor (0.779). The shuffled-text control is slightly negative (−0.062), so reading the correct text is doing real work in this environment — but not enough work to beat a text-free policy that was allowed a single tuned parameter.

7 Transfer and boundaries

Experiment B perturbs one environment factor at a time around the supply-side setting. We report *every* variant, including the ones that go the wrong way, and separate two reasons a normalised value can be negative.

Half the variants have no well-defined information value. In three of the six variants — long lead, lost sales and short lead — the oracle-belief reference itself

Table 5. Experiment B. *Every* one-factor-at-a-time variant, including the adverse ones. Values are the balanced weighted return J_w ; OIV is the oracle information value. $\dagger$ marks a variant whose OIV is not positive: there the oracle-belief reference itself underperforms the uninformed prior, so a normalised ratio cannot be read as recovery of information value and the finding is about controller–environment mismatch, not semantic sensing.

Variant	NoInfo	No-text	Rule	TF-IDF	GPT-4o	Oracle	OIV
baseline	1324	1713	1436	1362	1362	1435	+112
high_noise	1313	1674	1417	1344	1346	1411	+97
long_lead $\dagger$	-876	-967	-902	-869	-863	-878	-3
lost_sales $\dagger$	1512	1439	1489	1503	1507	1490	-22
low_noise	1334	1732	1449	1375	1372	1450	+116
short_lead $\dagger$	2398	2486	2411	2248	2213	2214	-183

underperforms the uninformed prior. The oracle information value is then negative, the normalised ratio’s denominator flips sign, and no statement of the form “this interpreter recovers x of the available information value” is meaningful. What those rows show is a fixed controller mismatched to a perturbed environment, not a failure of semantic sensing, and we mark them rather than reporting a ratio. This is a limitation of the diagnostic and of the fixed-controller design, and it is the reason raw paired effects are primary throughout this paper.

Interpreters can still lose badly. Under the short-lead variant of Table 5 the calibrated TF-IDF interpreter returns 2248 and GPT-4o 2213 against the uninformed prior’s 2398, so both lose well over a hundred reward. Halving the lead time leaves the controller with little time to act on a warning, so a belief that shifts stock upward mostly buys holding cost. An earlier version of this work omitted the short-lead variant and misreported the sign of the lost-sales case; both are now in the table, with every other variant.

Text–world pairing. The boundary study iterates over sixteen report templates, six of which describe normal operation. Simulating all of them in a disrupted world turns those six into false-negative reports. That is a legitimate experiment, but it is an evidence intervention, not mechanical robustness, and it breaks the text–world relationship the transfer experiment establishes. The default pairing now simulates each report in the world it describes; the false-negative arm remains available and is reported separately when used. Repairing the pairing changes conclusions: under the previous design the short-lead variant appeared to have a positive oracle information value, and it does not.

The text-free control again. In five of six variants the development-tuned no-text control returns more than every interpreter, and in the sixth the ordering is unchanged from the baseline. The controls are not a robustness afterthought; they are the comparison that determines what the interpreters have shown.

8 Limitations

The report feed and the environments are synthetic, single-SKU and finite-regime. Experiment R’s pools are generated rather than authored, so it carries no language-axis variance; its query is fixed and its relevance is defined by construction rather than by assessors. The dense retriever is one general-purpose sentence encoder with frozen embeddings, not a trained dense retrieval system, so we make no claim about the ceiling of dense retrieval here. The controller is fixed and not jointly optimised with the interpreter, so results describe value under a declared belief-to-action mapping rather than under every controller. Cached LLM outputs cover a small set of models with reconstructed rather than certified provenance. Evidence is consumed by concatenation; belief-level pooling over separately interpreted documents is a natural variant we do not evaluate.

9 Conclusion

Information access systems that feed decision policies should be evaluated on the decisions. In this benchmark, retrieval quality and sequential decision utility do not induce the same ordering, and the gap has a concrete mechanism: the ways a document can be non-relevant differ in how far they move a controller from a good action. The controls matter as much as the systems: a constant belief tuned only on development seeds is a demanding baseline, and effects measured against an uninformed prior alone can survive it or not. We release the benchmark, its frozen artifacts and its controls so that consequence-aware claims about evidence selection can be checked rather than asserted.

A Design, accounting and reproduction

Table 6. Experiment accounting, generated from the saved episode files rather than from design intent. ‘Episodes’ counts rows across all information sources; the number of *paired worlds* is seeds $\times$ templates, which is not the episode count. Experiment R draws a generated candidate pool per seed and so has no template axis.

Experiment	Seeds	Templates	Sources	Episodes
R (retrieval)	30	n/a	6	3,240
C (controlled)	20	36	12	8,640
M (multi-echelon)	30	24	12	8,640
S (supply-side)	30	16	12	5,760

No-text operating point. Experiment M selected $p_{\text{event}} = 1.0$ on seeds 3200–3229, Experiment S selected $p_{\text{event}} = 1.0$ on seeds 4200–4229. Both development seed ranges are disjoint from every confirmation seed range, and selection used the balanced estimand. The arm is tuned, but never on its own evaluation data.

Environment, costs and timing. Single-item lost-sales inventory over a finite horizon of $T = 40$ periods: revenue 10 per unit sold, ordering cost 5 fixed plus 2 per unit, holding cost 1 per unit of ending inventory, stockout penalty 8 per unit of unmet demand, initial on-hand 30, Poisson demand with base mean 8. Three regimes: NORMAL; SUPPLIER DELAY, raising the lead time during the event window; and DEMAND SURGE, multiplying the demand rate. The supply-side transfer instead reduces a factory’s capacity in a multi-echelon network. Within a period the order is: receive arrivals, realise demand, fulfil from on-hand, place an order. An order placed in period t under lead time L arrives at the start of period $t + L$; orders already in transit keep their arrival period, so a lead-time change applies only to orders placed from that period onward. The simulator and every planner share one implementation of this schedule, so a planner’s arrival model cannot drift from the dynamics it plans against.

Information release. Warning text is published at period 12; the event window begins at period 20. Before publication no interpreter has read anything and every controller plans under the benchmark prior. From publication the interpreted belief replaces the prior while on-hand stock and the in-transit pipeline carry over unchanged.

Belief to action. The controlled experiments solve a receding-horizon linear program each period and execute only its first order, with belief-weighted expected lead time and demand per future period. The multi-echelon experiments use a belief-adaptive base-stock policy with effective rate $\mathbb{E}_b[\mu]$ and order-up-to target $\mathbb{E}_b[\mu]\bar{L} + 1.65\sqrt{\bar{L}}\mathbb{E}_b[\mu]$. The controller is identical across information sources within an experiment.

Interpreters, information and retrieval. No non-oracle interpreter receives the regime, future demand, future state or environment internals; the oracle-belief reference receives the true regime and is a reference, not a deployable system. The TF-IDF classifiers are fitted only on training templates and evaluated on disjoint held-out ones. LLM beliefs are replayed from frozen canonical responses, with model labels reconstructed by re-deriving each cache key from every known (model, text) pair and unmatched entries labelled **unknown**. BM25 uses the Okapi form with $k_1 = 1.5$, $b = 0.75$ and idf floored at zero; dense retrieval is cosine similarity over frozen **all-MiniLM-L6-v2** embeddings, shipped so the published numbers reproduce without model weights. Ties break by document id.

Reproduction and impact. The release reconstructs its own offline inputs: cached semantic responses come from a tracked, checksum-verified manifest, and classifier checkpoints are retrained from tracked templates with fixed seeds; neither step needs credentials or network access. A verification entry point replays stored beliefs through the simulator and regenerates every table here from the frozen episodes. This paper shows that a decision-support component can look better on upstream metrics while being worse to act on, so deployment needs human oversight and conservative fallbacks; a synthetic study does not establish deployment safety.

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